ITA06020 MACHINE LEARNING

Experiment 15 – Iris Flower Classification Using Naive Bayes

Aim:
To classify Iris flowers into different species using the Naive Bayes classification algorithm.

Procedure:

1. Import the required Python libraries.
2. Load the Iris dataset.
3. Divide the dataset into training and testing data.
4. Train the Gaussian Naive Bayes classifier.
5. Predict the flower species and calculate accuracy.

Dataset: Iris Flower Dataset (built-in Scikit-learn dataset).

Python Code:

```
from sklearn.datasets import load_iris

from sklearn.model_selection import train_test_split

from sklearn.naive_bayes import GaussianNB

from sklearn.metrics import accuracy_score, confusion_matrix

iris = load_iris()

X = iris.data

y = iris.target

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42, stratify=y

)

model = GaussianNB()

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

sample = [[5.1, 3.5, 1.4, 0.2]]
```

```
prediction = model.predict(sample)

print("Predicted Flower:", iris.target_names[prediction[0]])

print("Accuracy:", accuracy * 100, "%")

print("Confusion Matrix:")

print(confusion_matrix(y_test, y_pred))
```

Output:

```
Predicted Flower: setosa
Accuracy: 96.66666666666667 %
Confusion Matrix:
[[10  0  0]
 [ 0  9  1]
 [ 0  0 10]]
```

Result:
The Iris flower classification was successfully performed using the Naive Bayes algorithm with an accuracy of 96.67%.